mRNA Folding

1. Context and Relevance

Understanding how a sequence of mRNA bases folds into its functional structure is a central challenge in molecular biology. The folding process determines how genetic information is translated and regulated, influencing everything from protein synthesis to gene expression control. Small variations in folding can alter how mRNA interacts with ribosomes and other cellular machinery, ultimately affecting which proteins are produced and in what quantities. Accurate models of mRNA folding are therefore essential not only for decoding the mechanisms of gene regulation but also for designing RNA-based therapies and vaccines that depend on predictable and stable secondary structures.

Predicting the folded conformation given a chain of mRNA bases is a computationally hard problem. In essence, it is a combinatorial optimization task. In a simplified model, thiank of the following: among many possible pairings and spatial arrangements of bases, the goal is to find the structure with the lowest free energy. Classical algorithms, such as dynamic programming or stochastic sampling, can approximate solutions for short sequences but scale poorly as the length of the mRNA sequences increase. Machine learning methods have improved prediction accuracy by learning statistical patterns from known structures, yet they remain limited by data availability and computational cost. Quantum computing offers a possible new avenue, as quantum algorithms may be able to explore vast energy landscapes more efficiently, potentially identifying low-energy fold conformations that are inaccessible to classical approaches.

2. Challenge Overview

In this hackathon challenge, participants are tasked with designing a computational pipeline which takes an unfolded mRNA sequence and predicts its lowest energy conformation (= lowest energy secondary structure, these are synonyms), from which its folded state can be derived. In industry such problems are tackled by a series of steps to break down the complexity of the problem to a handleable format for a classical or quantum computer. Such a pipeline consists of classical data preprocessing, the main quantum program given by a quantum sampler, and classical postprocessing. We will follow the same approach here. In

particular, the quantum computing part of the pipeline will be a sampler for possible low-energy conformations that have to be (classically) post-processed.
Specifically, participants are encouraged to:

- Start with a quantum sampler that outputs options for mRNA foldings based on rewarding certain substructures like binding base pairs or quartets over unbinding ones. Use a simple quantum encoding that outputs solutions for very small short mRNA chains, before moving on to a more complicated model. See below for a simple encoding strategy.
- Think about the classical preprocessing necessary for the chosen encoding strategy and implement this, too.
- Take the output of the sampler and apply classical postprocessing: First the output of the quantum circuit has to be checked for validity (typically not all predicted conformations are physically possible). Secondly, the output is fed into a provided tool which computes the total energy of a given conformation (tool can be found in Resources below).
- Run the pipeline iteratively to eventually aggregate conformations of lower and lower energy until the lowest energetic conformation has been found.
- Enhance your workflow by making use of provided information of known secondary substructures for a given mRNA chain as classical preprocessing to reduce the problem size for the quantum sampler.
- Clearly document your workflow to ensure reproducibility and extensibility
- Demonstrate a compelling path toward a potential quantum advantage, with a clear articulation of the challenges that must be overcome.
- If time allows: Design a more sophisticated classical preprocessing step which reduces the problem size further and allows scaling the pipeline up to longer mRNA chains.

3. Detailed Problem Description

a. Mathematical Problem / Model Description

The goal of the challenge is to design a computational pipeline which, for a given mRNA chain, finds its lowest energy conformation for a given input sequence.

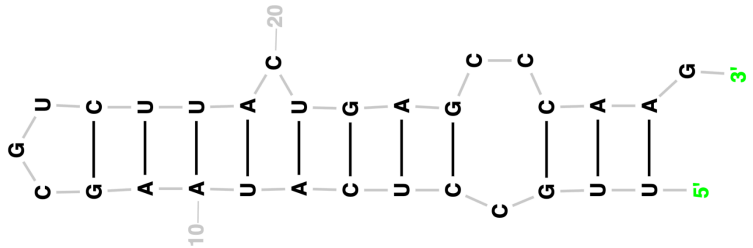
mRNA are sequences of four nitrogenous bases, adenine (A), uracil (U), cytosine (C) and guanine (G). The sequence is given in a single letter notation, for example:

UUGCCUCAUAAGCGUCUUACUGAGCCCAAG

Here, each entry represents one nitrogenous base starting from one end of the mRNA chain. The four bases can bind strongly for the specific basepairs **A-U** and **G-C.** Other base pairings are possible, e.g. **G-U** or **G-G**, but are less energetically favorable.
These pairings are fundamental for forming the mRNA's double-stranded regions and are guided by the ability of the bases to form hydrogen bonds with each other. There is an exponentially growing search space of possible combinations with increasing chain length. The different folded secondary structures are expressed in a dot-bracket notation which describes the mRNA's secondary structure:

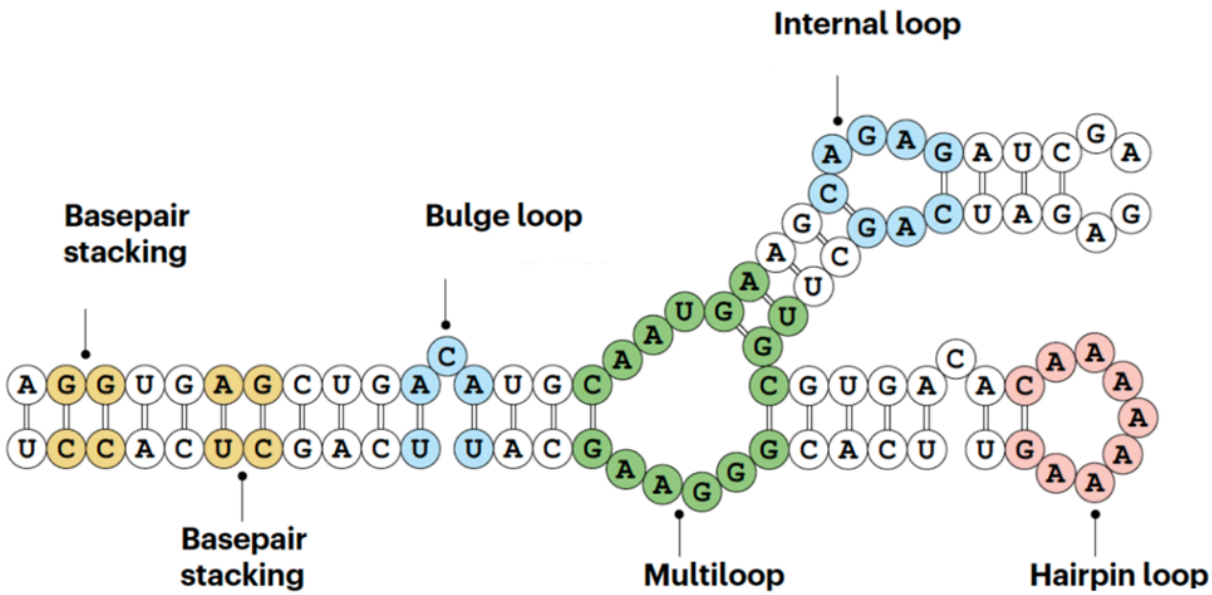
(((.(((((((...)))).)))).)).))).

Open parentheses indicate that the base is paired to another base ahead of it. Closed parentheses indicate that a base is paired to another base behind it. Periods indicate an unpaired base. The number of open and closed parentheses will always be equal. For the above example the folded conformation in 2D would look like this:



where the chain starts at the bottom left and the numbers 10 and 20 refer to the base index along the chain.

Another, more complicated example of a folded mRNA structure can be found below.



For quartets like GCGC or AUGC there are energetic rewards (meaning that the free energy is smaller with these quartets, while there are energetic penalties for bulge loops, multiploops and internal loops or hairpins. These penalties and rewards are crucial for guiding the optimization to the minimal energy secondary structure and for calculating the energy of the fully folded sequence. A database for rewards and penalties can be found here: https://rna.urmc.rochester.edu/NNDB/ (also linked in the resources below).

One way to mathematically formulate the problem in terms of an optimization problem is the following: The idea is to phrase the problem as a quadratic unconstrained binary optimization problem (QUBO). For this, let us number each position of a base in the mRNA sequence of total length n by integers 1,...,n. Now, instead of only looking for pairs of bases we consider quartets. A possible quartet is two base pairs in a row that match, i.e. position k & l match next to positions k+1 & l-1, which also match. Let us now consider all possible quartets of this type, summarize them in the set Q, and use the label q_i for the ith quartet in this set, s.t. $q_i=1$ if this quartet is realized in the folded mRNA and $q_i=0$ otherwise (i.e. q_i are binary variables). The set Q must be determined in classical preprocessing.

Eventually we want to minimize the free energy of the total folding structure, and this can be approximated by summing up the energetic rewards and penalties for certain structures to be realized. Besides the energetic rewards for an individually realized quartet, there is an additional reward for two consecutive quartets being realized. For a quartet q_i let us define the set $QS(q_i)$ as the set of quartets that can be stacked with quartet q_i. Then such reward is realized iff $q_iq_j=1$ for $q_j \in QS(q_i)$. Just like Q the set $QS(q_i)$ must be determined in classical preprocessing for each possible quartet q_i.

Clearly, a base that is used in one realized quartet cannot be used again. Hence, let us define QC to be the set of crossing pairs of quartets, the constraint then being $q_i q_j = 0$ for $(q_i, q_j) \in QC$. Again, QC needs to be determined in classical preprocessing.
In summary, in its simplest form, this leads to the following objective function

$$\min \sum_{q_i \in Q} e_{q_i} q_i + r \sum_{q_i \in Q} \sum_{q_j \in QS(q_i)} q_i q_j$$

where e_{qi} is the free energy gained by quartet q_i being realized in the folded sequence. r is the energetic reward for two consecutive quartets being realized, which is made independent from the actual base pairs that make up the consecutively paired quartets for simplicity. This is all subject to the constraint $q_i q_j = 0$ for $(q_i, q_j) \in QC$, which must be incorporated into the objective function with an appropriate penalty term. It is left to the teams how to do so.

Importantly: The parameter r as well as an additional parameter tuning the penalty term for the constraint have to be played with. Using a quantum computer (or to start with, a simulator) to sample from the possible solution space to this QUBO, one can find good values for these parameters before moving on to integrating everything in one pipeline.

Alternative formulations and extensions:
In a slightly more complicated form of the problem, an additional penalty term could be introduced to penalize stacked realized quartets that end with **A-U**. We leave it to the teams to think about how this would change the above objective function and what additional preprocessing would be necessary to incorporate such a term.
Also, one is not limited to applying this QUBO formulation to quartets. One could also consider 2x3 or 2x4 packs of base pairs and implement the QUBO for longer paired subsequences to increase the length of mRNA chains which be studied.

Classical Pre- and Post-Processing:

In order to obtain all the sets of all quartets Q and quartet sequences QS one first has to apply classical pre-processing to create a combination matrix of all possible pair combinations with. Here each possible/impossible pair combination is marked by one and zero respectively. Each quartet or longer sequence then corresponds to a diagonal of ones from top right to left. The matrix usually looks like this (diagonals are marked in pink).

	G	G	A	A	G	C	A	A	A	C	A	U	C	C	C	U	G	U
G		0	0	0	0	1	0	0	0	1	0	1	1	1	1	1	0	1
G			0	0	0	1	0	0	0	1	0	1	1	1	1	1	0	1
A				0	0	0	0	0	0	0	0	1	0	0	0	1	0	1
A					0	0	0	0	0	0	0	1	0	0	0	1	0	1
G						1	0	0	0	1	0	1	1	1	1	1	0	1
C							0	0	0	0	0	0	0	0	0	0	1	0
A								0	0	0	0	1	0	0	0	1	0	1
A									0	0	0	1	0	0	0	1	0	1
A										0	0	1	0	0	0	1	0	1
C											0	0	0	0	0	0	1	0
A												1	0	0	0	1	0	1
U													0	0	0	0	1	0
C														0	0	0	1	0
C															0	0	1	0
C																0	1	0
U																	1	0
G																		1
U																		

As a first step use the diagonals of length maximum length 2 and assign each of them to a qubit q_i as described above. After a first working prototype has been set up, alternative encodings can also be used to define the cost function, like classical pre-selecting longer pair sequences as those are usually more stable and are present in the final folded conformation. Alternatively, longer encodings sequences could be encoded in the quantum layer instead of quartets.

Besides the classical pre-processing, post-processing is also needed to only obtain valid conformations. The first step includes checking that no quartets are overlapping. Afterwards, the quartet representation has to be rewritten into a base pair sequence in dot-bracket representation. Afterwards the valid conformation is fed into the provided RNA package tool to compute its total energy (see resources below).

This output is then fed back into the quantum circuit to converge towards lower energy conformations.

b. Proposed Quantum Approaches

Quantum computing approaches for this challenge should take sequences of base pairs of moderate lengths and output candidates of foldings in the dot-bracket notation introduced above. In essence, the quantum computing subroutine should be used as a sampler after classical preprocessing and followed by classical postprocessing. This means that the cost function only functions as a guide towards lower energy conformation but will not necessarily converge to the absolute minimum. Classical postprocessing must be applied to make sure that the candidates are indeed possible foldings (as, depending on the approach taken, outputs are not necessarily guaranteed to fulfil this). This can be done by applying and/or modifying the above encoding, or by considering a novel approach to the problem.
Suggested quantum computing approaches include:

- Variational quantum algorithms, such as QAOA or VQE, to minimize the objective function.
- Variants of other quantum optimisation algorithms.

Both suggested approaches directly relate to finding extrema of the objective function described above in the mathematical problem description.

The participants are encouraged to also think about fault-tolerant quantum algorithms that may further enhance the computational pipeline. However, when it comes to the working prototype, a key deliverable of this hackathon challenge, NISQ algorithms will likely be the only viable option.

4. Expected Deliverables

- **Working Prototype:** A functional model that takes data as input and produces a solution to the posed challenge. We recommend a step-by-step approach, beginning with a proof-of-concept for a very simple toy model before expanding to a full model, e.g. tackling larger data sets or problem instances.
- **Technical Explanation:** Participants must be able to explain the quantum algorithms used, the data model, and the rationale for choosing the specific approach. The focus should be on the clear and deliberate use of methodology and the benchmarking process, rather than solely on high-performing results. Participants should benchmark their quantum model against standard classical solutions. If more than one model was developed during the hackathon the team may also benchmark against those.
- **Classical Bottleneck Analysis and Justification for Quantum Computing**

Approach: An analysis of standard approaches and their bottlenecks. This is the basis for arguing why a quantum computing approach may be beneficial for the problem at hand.

- **SDG Impact Assessment:** An assessment of the impact of your solution on the main SDG tackled by this challenge. Your arguments should be backed by references to publications and publicly available data.
  Also: What other SDGs may be impacted by your solution (both to the positive or to the negative)? List them and argue for why your solution is relevant also to those.
- **Business Case:** A concise plan outlining the commercial viability, target users, and market strategy. Think about how your solution could be commercialized, how you would approach commercialization, what would the product be, who would be the customers, how you would make money with it.
  Also: In what other fields / problems could your solution be useful?
- **Pitches & Technical Deep Dive:** At the end of the hackathon you will have the chance to present your solution in a final presentation as a 3 min pitch with slides. Please explain the solution's potential and its future roadmap, including an honest discussion of the challenges that need to be addressed before it can be a fully realized product, all based on the above deliverables. The pitch will be followed by a 2 min Q&A with the jury for the respective challenge, which consists of 50% technical experts and 50% business experts.
  In addition, you will have the chance to present your solution in a 10 min technical deep dive to the technical experts in the jury. For this, no slides are required, but a prepared code review is envisioned.

Importantly: A deeper investigation into one quantum computing approach is preferred over a superficial overview of many. The goal is not to find a single "best" model, but to demonstrate a clear understanding of the chosen approach. The solution should focus on demonstrating how the quantum approach can be applied to the problem, specifically addressing the challenge specified above. The goal is to articulate a compelling case for a future computational speed-up or greater accuracy, rather than proving an immediate advantage.

Target audience: The project should be compelling to technical professionals who are interested in exploring how quantum computing can be applied to their domain.

5. Resources

[important references]

[Technical Paper] mRNA secondary structure prediction with quantum computers:

https://arxiv.org/abs/2405.20328

[Technical Paper] QUBO for RNA folding problem:
https://arxiv.org/abs/2208.04367

[Tool] Given mRNA sequence and its folding in the dot-bracket notations, this tool visualizes the folding, incl. documentation: https://rnacentral.org/r2dt
For a start, try this command, which corresponds to the example in the text above:
>mRNA_40
UUGCCUCAUAAGCGUCUUACUGAGCCCAAG
(((.(((((((...)))).)))).)..))).

[Tool] RNA package for calculating the conformation energy:
ttps://viennarna-python.readthedocs.io/en/master/
For a start, try this command:

```
import RNA
# Input RNA sequence
sequence = "GGAGCAAAACUUGUCGAUUGAGAACAAAAUACAGAAUUUGCUUG"
# Secondary structure (in the dot-bracket notation,
see https://www.tbi.univie.ac.at/RNA/ViennaRNA/refman/io/rna_structures.html#dot-bracket-notation)
structure = '................(((....)))................'
# Compute the energy of the specified folding for the sequence
RNA.fold_compound(sequence).eval_structure(structure)
```

[Database] Nearest-Neighbor Database for minimal energy calculations:
https://rna.urmc.rochester.edu/NNDB/
Particularly, this dataset contains all energy rewards and penalties of quartets:
https://rna.urmc.rochester.edu/NNDB/parameter_tables/turner_2004_rnastructure/turner_2004_watson_crick_stack_dg.txt